

Mediterranean diet, stroke, cognitive impairment, and depression: A meta-analysis.

OBJECTIVE: This meta-analysis aims to quantitatively synthesize all studies that examine the association between adherence to a Mediterranean diet and risk of stroke, depression, cognitive impairment, and Parkinson disease. **METHODS:** Potentially eligible publications were those providing effect estimates of relative risk (RR) for the association between Mediterranean diet and the aforementioned outcomes. Studies were sought in PubMed up to October 31, 2012. Maximally adjusted effect estimates were extracted; separate analyses were performed for high and moderate adherence. **RESULTS:** Twenty-two eligible studies were included (11 covered stroke, 9 covered depression, and 8 covered cognitive impairment; only 1 pertained to Parkinson's disease). High adherence to Mediterranean diet was consistently associated with reduced risk for stroke (RR = 0.71, 95% confidence interval [CI] = 0.57-0.89), depression (RR = 0.68, 95% CI = 0.54-0.86), and cognitive impairment (RR = 0.60, 95% CI = 0.43-0.83). Moderate adherence was similarly associated with reduced risk for depression and cognitive impairment, whereas the protective trend concerning stroke was only marginal. Subgroup analyses highlighted the protective actions of high adherence in terms of reduced risk for ischemic stroke, mild cognitive impairment, dementia, and particularly Alzheimer disease. Meta-regression analysis indicated that the protective effects of Mediterranean diet in stroke prevention seemed more sizeable among males. Concerning depression, the protective effects of high adherence seemed independent of age, whereas the favorable actions of moderate adherence seemed to fade away with more advanced age. **INTERPRETATION:** Adherence to a Mediterranean diet may contribute to the prevention of a series of brain diseases; this may be of special value given the aging of Western societies. © 2013 American Neurological Association.